

CODECHECK certificate 2026-018



WARNING

Report DOI is a placeholder: **NOT SET**
This certificate is not yet finalized.
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valid values before finalizing.

Item	Value
Title of checked publication	Improving reproducibility of GIScience publications through novel reproducibility guidelines and revised review procedures
Author(s)	Carlos Granell, Frank O. Ostermann, Daniel Nüst, Peter Kedron, Eftychia Koukouraki, Miguel Matey-Sanz, Rémy Decoupes, Sergio Trilles, Anita Graser, Tom Niers
Reference	https://doi.org/10.5311/JOSIS.2026.32.591
Codechecker(s)	Alexander Trinidad
Date of check	2026-07-17 11:50:00
Summary	Quarto and Jupyter notebooks used to reproduce the results reported by Granell et al. (2026) and evaluate the reproducibility of the research outputs.
Repository	https://github.com/alexandertrinidad/reproducible-research-giscience-longitudinal-study

Table 1: CODECHECK summary

Output	Comment	Size (b)
outputs/AGILE_pre_post.png	manuscript Figure 1	
outputs/AGILE_sankey_year.png	manuscript Figure 2	
outputs/figure2_table.png	manuscript table under Figure 2	
outputs/GIScience_pre_post.png	manuscript Figure 3	
outputs/table1.png	manuscript Table 1	
outputs/table5.png	manuscript Table 5.	
outputs/table6.png	manuscript Table 6.	
outputs/table7_1.png	manuscript Table 7 (first part)	
outputs/table7_2.png	manuscript Table 7 (second part)	

Table 2: Summary of output files generated

Summary

This code was straightforward to codecheck.

The CODECHECK report documents the reproduction of “Improving reproducibility of GIScience publications through novel reproducibility guidelines and revised review procedures”, by Carlos Granell and colleagues, published in the Journal of Spatial Information Science (JOSIS), DOI: 10.5311/JOSIS.2026.32.591. The computational workflow was reproduced in a local environment, and the reproduction materials and generated outputs were uploaded to a dedicated repository. All provided Quarto and Jupyter notebooks executed successfully without major computational issues. The reproduced analyses generated all substantive results, figures, and tables reported in the published paper. The numbers of total and eligible papers in the analysed corpus matched the corresponding numbers in the paper. The descriptive results for the AGILE and GIScience conference papers, the comparisons between conferences, the hypothesis tests, the assessment-process analyses, and the authorship analyses were also successfully reproduced. Where numbering differed between the manuscript and the rendered notebooks, the underlying values and conclusions remained consistent. All figures were reproduced exactly, although their numbering sometimes differed between the paper and the computational outputs. Figure 1 in the paper corresponded to Figure 4 in the rendered results notebook, Figure 2 corresponded to Figure 1 in the discussion notebook, and Figure 3 was reproduced as Figure 3 in the relevant rendered notebook. The pairwise group comparisons and the results discussed on pages 76–83 of the paper were likewise supported by the reproduced notebook outputs. The reproduction instructions in the README file were clear for executing the workflow locally.

Some table cross-references in the Quarto notebooks and the CLAUDE.md file correspond to the numbering used in the preprint rather than the final published version. For example, a reproduced table described as corresponding to Table 3 actually corresponds to Table 6 in the published paper, while another described as corresponding to Table 4 corresponds to Table 7.

These discrepancies relate only to table labelling and cross-referencing, not to the computational results. Overall, all results, figures, and tables reported in the published paper were successfully reproduced.

This report includes only the figures and tables from the paper. For all other results, please refer to the rendered Quarto HTML outputs in the codecheck/outputs folder.

CODECHECKER notes

The reproduction was carried out on a MacBook Pro equipped with an Apple M5 processor, 32 GB of memory, and macOS Tahoe 26.5.2. The authors' repository was obtained from: <https://github.com/nuest/reproducible-research-giscience-longitudinal-study>

The computational workflow included code written in both R and Python. The reproduction followed these steps:

1. Cloned the authors' reproducibility repository from GitHub.
2. Executed the Quarto and Jupyter notebooks sequentially, starting with notebook 01 and continuing through notebook 07.

No major problems were encountered while running the workflow. The entire CODECHECK process took approximately 60 minutes. This duration includes the additional time required because this was my first time conducting a CODECHECK. The computational reproduction itself took less time, but I am unable to provide a reliable estimate of the execution time alone.

Recommendations

No recommendations. The code run smoothly.

Manifest files

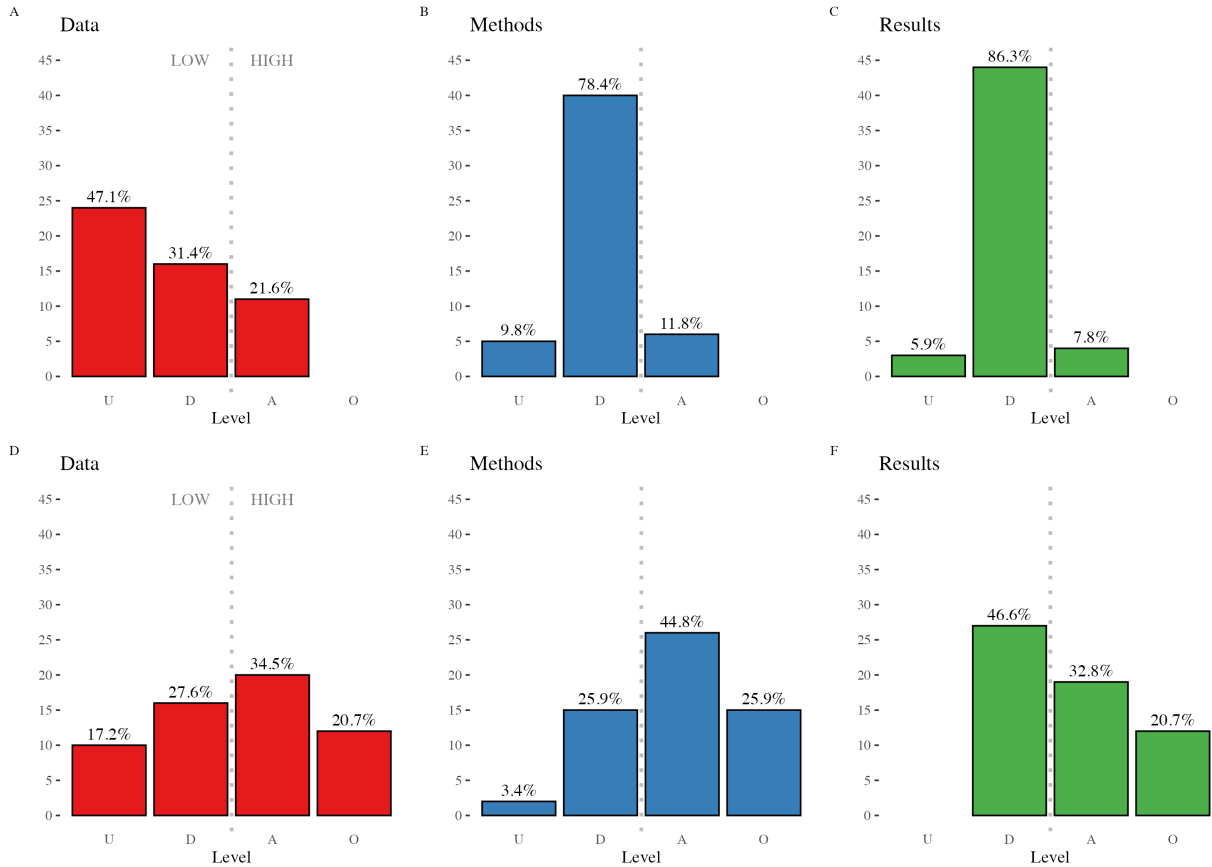
AGILE_pre_post.png

Comment: manuscript Figure 1

AGILE paper assessment according to the criteria 'Data', 'Methods', and 'Results'.

Top: pre-intervention (2017-2019) papers (N=51). Bottom: post-intervention (2021-2024) papers (N=58).

X-axis labels: (U)ndocumented; (D)ocumented; Documented and (A)vailable; Documented, Available and (O)pen.



Note: AGILE 2020 papers excluded (transition year).

AGILE_sankey_year.png

Comment: manuscript Figure 2

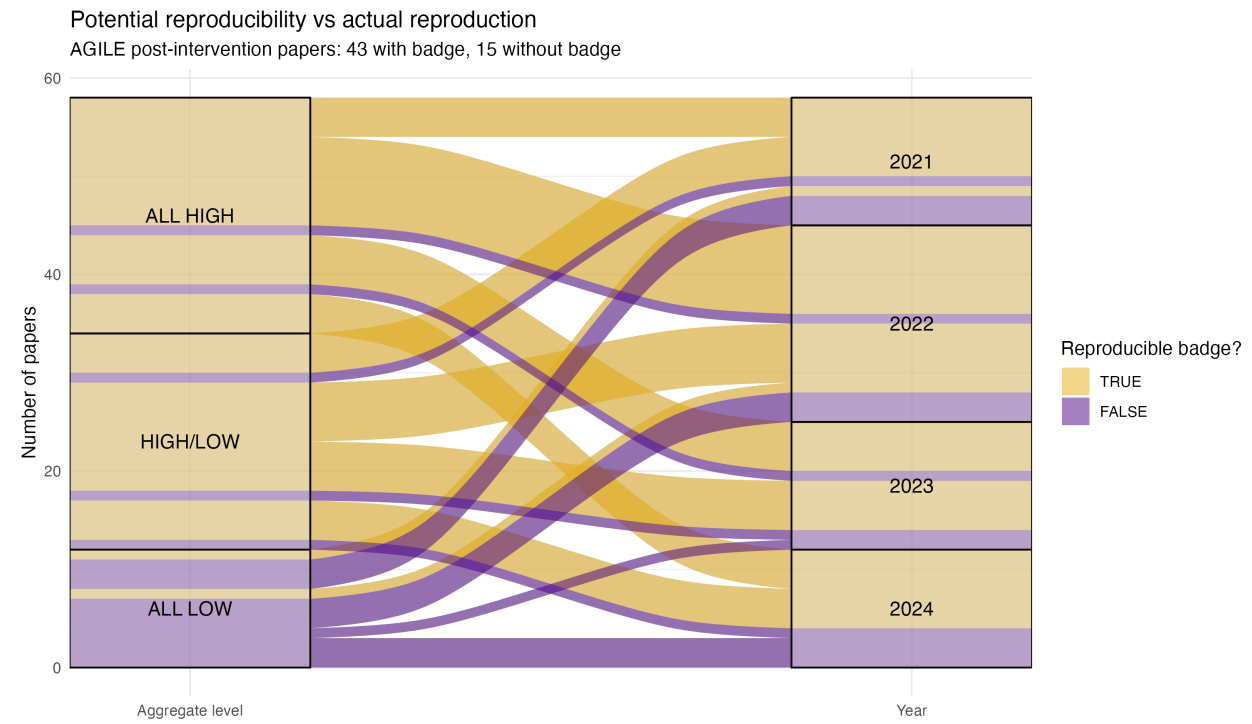


figure2_table.png

Comment: manuscript table under Figure 2

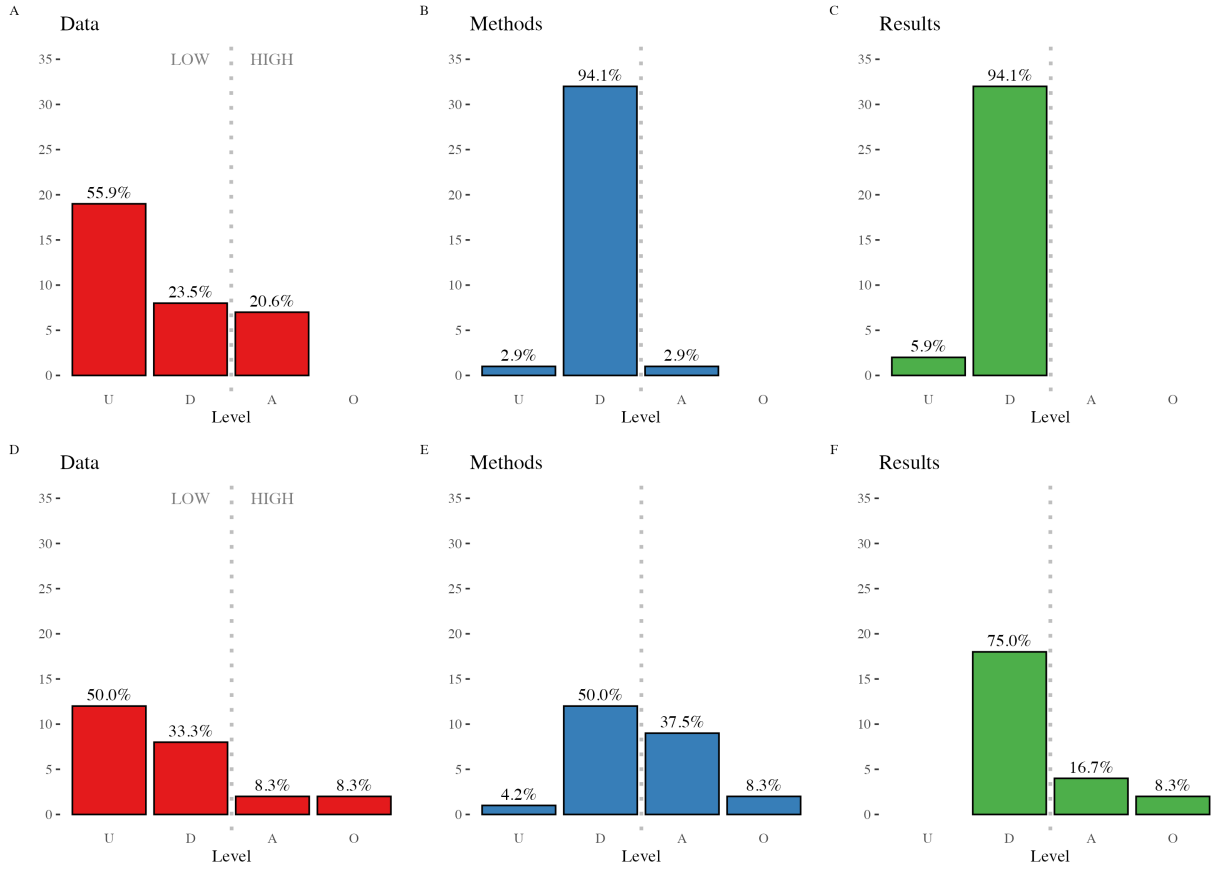
agg_level	n	proportion
TRUE		
ALL HIGH	22	51.2
HIGH/LOW	19	44.2
ALL LOW	2	4.7
FALSE		
ALL HIGH	2	13.3
HIGH/LOW	3	20.0
ALL LOW	10	66.7

Comment: manuscript Figure 3

GIScience paper assessment according to the criteria 'Data', 'Methods', and 'Results'.

Top: pre-guidelines (2016, 2018) papers (N=34). Bottom: post-guidelines (2021 Part II, 2023) papers (N=24).

X-axis labels: (U)ndocumented; (D)ocumented; Documented and (A)vailable; Documented, Available and (O)pen.



Note: GIScience 2021 Part I papers excluded (transition year).

table1.png

Comment: manuscript Table 1

Table 1: Distribution of eligible conference papers

Distribution of eligible conference papers (N=167)

Table 1 of the published paper - Note: transition year is excluded here

	Year	no. papers	no. papers eligible
agile			
	2017	20	16
	2018	19	17
	2019	19	18
	2020	22	0
	2021	16	13
	2022	22	20
	2023	14	13
	2024	14	12
Total	—	146	109
giscience			
	2016	21	17
	2018	17	17
	2020	16	0
	2021	13	13
	2023	11	11
Total	—	78	58

table5.png

Comment: manuscript Table 5.

GIScience vs AGILE: Absolute change in percentage points		
Variation between <i>HIGH</i> and <i>LOW</i> before and after the intervention.		
Aggregate level per criterion*	<i>GIScience</i>	<i>AGILE</i>
Input data-HIGH	−3.9%	33.6%
Methods-HIGH	42.9%	58.9%
Results-HIGH	25.0%	45.6%

table6.png

Comment: manuscript Table 6.

Table 1: Distribution of disagreement types

Distribution of disagreement types			
No grouping (N=224). Table 3 of the published paper			
	Disagreement type	#	%
	uncertain assessment	100	44.6%
	no disagreement	63	28.1%
	significant disagreement	42	18.8%
	borderline conceptual paper	14	6.2%
	annotation inconsistencies	5	2.2%
Total	—	224	1

table7_1.png

Comment: manuscript Table 7 (first part)

Table 5: Distribution of 'Uncertain disagreement'



Uncertain disagreement (N=100)									
Applied to <i>Input data</i> , <i>Methods</i> , ... and <i>Results</i> dimensions.									
<i>Conf Series</i>	Input Data			Methods, ...			Results		
	<i>Disagr?</i>	#	%	<i>Disagr?</i>	#	%	<i>Disagr?</i>	#	%
AGILE	no	23	35.4%	no	46	70.8%	no	36	55.4%
AGILE	yes	42	64.6% ¹	yes	19	29.2% ²	yes	29	44.6%
GISCIENCE	no	7	20.0%	no	24	68.6%	no	28	80.0%
GISCIENCE	yes	28	80.0% ¹	yes	11	31.4%	yes	7	20.0% ³
¹ Major disagreements at both conferences.									
² Minor disagreements at the AGILE conference.									
³ Minor disagreements at the GIScience conference.									

table7_2.png

Comment: manuscript Table 7 (second part)

Table 6: Distribution of 'significant disagreement'

Significant disagreement (N=42)									
Applied to <i>Input data</i> , <i>Methods</i> ,... and <i>Results</i> dimensions.									
<i>Conf Series</i>	Input data			Methods,...			Results		
	<i>Disagr?</i>	#	%	<i>Disagr?</i>	#	%	<i>Disagr?</i>	#	%
AGILE	no	6	20.0%	no	9	30.0%	no	13	43.3%
AGILE	yes	24	80.0% ¹	yes	21	70.0%	yes	17	56.7% ²
GISCIENCE	no	3	25.0%	no	5	41.7%	no	5	41.7%
GISCIENCE	yes	9	75.0% ¹	yes	7	58.3% ³	yes	7	58.3% ³

¹ Major disagreements at both conferences.

² Minor disagreements at the AGILE conference.

³ Minor disagreements at the GIScience conference.

Acknowledgements

I would like to thank Dr. Nüst for introducing me to the CODECHECK workflow and the other authors for providing the reproducible workflow accompanying their paper. CODECHECK is financially supported by the Mozilla foundation.

Citing this document

About CODECHECK

This certificate confirms that the codechecker could independently reproduce the results of a computational analysis given the data and code from a third party. A CODECHECK does not check whether the original computation analysis is correct. However, as all materials required for the reproduction are freely available by following the links in this document, the reader can then study for themselves the code and data.

About this document

This document was created using R Markdown using the `codecheck` R package. `make codecheck.pdf` will regenerate the report file.

```
sessionInfo()
```

```
## R version 4.6.0 (2026-04-24)
## Platform: aarch64-apple-darwin23
## Running under: macOS Tahoe 26.5.2
##
## Matrix products: default
## BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/
## LAPACK: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRlapack.dylib; LAPACK version 3.12.0
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/Berlin
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets
## [6] methods    base
##
## other attached packages:
## [1] readr_2.2.0      tibble_3.3.1
## [3] xtable_1.8-8     yaml_2.3.12
## [5] rprojroot_2.1.1  knitr_1.51
## [7] codecheck_0.25.0.9000 parsedate_1.3.2
## [9] R.cache_0.17.0   gh_1.6.0
##
## loaded via a namespace (and not attached):
## [1] xfun_0.60      httr2_1.3.0      tzdb_0.5.0
## [4] vctrs_0.7.3    tools_4.6.0      generics_0.1.4
## [7] curl_7.1.0     pkgconfig_2.0.3  pdftools_3.9.0
## [10] R.oo_1.27.1    readxl_1.5.0     assertthat_0.2.1
## [13] lifecycle_1.0.5 compiler_4.6.0    stringr_1.6.0
## [16] git2r_0.36.2   tinytex_0.60     keyring_1.4.1
```

## [19]	httpuv_1.6.17	htmltools_0.5.9	later_1.4.8
## [22]	pillar_1.11.1	whisker_0.4.1	R.utils_2.13.0
## [25]	openssl_2.4.2	cachem_1.1.0	zen4R_0.10.6
## [28]	magick_2.9.1	tidyselect_1.2.1	digest_0.6.39
## [31]	stringi_1.8.7	dplyr_1.2.1	purrr_1.2.2
## [34]	rsvg_2.7.0	fastmap_1.2.0	cli_3.6.6
## [37]	magrittr_2.0.5	triebeard_0.4.1	XML_3.99-0.23
## [40]	crul_1.6.0	utf8_1.2.6	osfr_0.2.9
## [43]	withr_3.0.3	promises_1.5.0	rmarkdown_2.31
## [46]	httr_1.4.8	gitcreds_0.1.2	otel_0.2.0
## [49]	qpdf_1.4.1	cellranger_1.1.0	askpass_1.2.1
## [52]	R.methodsS3_1.8.2	hms_1.1.4	memoise_2.0.1
## [55]	evaluate_1.0.5	urltools_1.7.3.1	rlang_1.3.0
## [58]	Rcpp_1.1.2	glue_1.8.1	httpcode_0.3.0
## [61]	xml2_1.6.0	fauxpas_0.6.0	rorcid_0.7.0
## [64]	jsonlite_2.0.0	R6_2.6.1	plyr_1.8.9
## [67]	fs_2.1.0		